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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

STANDARD OPERATING MANUAL: CONCENTRATION & DECOMPOSITION SECTION

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SECTION 1: PROCESS DESCRIPTION & NORMAL OPERATING ENVELOPE

1.1 Preflash Concentration Section (V-2301, P-2301A/B, X-2301):

- Dilute 28 wt% CHP feed (Stream S210, 42,500 kg/hr) is concentrated under 35 mmHgA vacuum in Preflash Column V-2301
- Safe Operating Limits for V-2301:
 - * Sump Temperature: Normal 92.0 degC | High Alarm 98.0 degC | High-High Trip TAAH-0402 = 105.0 degC (Interlock)
 - * Overhead Vacuum: Normal 35.0 mmHgA | High Alarm 55.0 mmHgA | High-High Trip PAHH-0401 = 75.0 mmHgA

1.2 Decomposer Reactor Section (D-2304, E-2307, P-2304A/B, D-2310):

- Concentrated 82 wt% CHP (Stream S229) is cleaved into equimolar Phenol and Acetone in Decomposer Reactor D-2304
- Because CHP cleavage is strongly exothermic (-1580 kJ/kg), D-2304 operates as a back-mixed loop reactor with a
- Safe Operating Limits for D-2304:
 - * Reactor Temperature: Normal 82.0 degC | High Alarm TAH-1301 = 92.0 degC | SIL-2 Trip TXSHH-1301 = 115.0 degC
 - * Reactor Pressure: Normal 1.8 kg/cm2g | High Alarm PAH-1301 = 3.2 kg/cm2g | SIL-2 Trip PAHH-1301 = 5.5 kg/cm2g
 - * Reactor Level (LT-1301): Normal 55.0% | Low-Low LSLL = 20.0% | High-High LSHH = 82.0%

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OM-2300 PAGE 2 OF 5 - SECTION 2: NORMAL STARTUP SEQUENCE (COLD INVENTORY ESTABLISHMENT)

Step 1: Verify Emergency Cumene Quench Drum is filled with ≥ 35 m3 chilled Cumene at 30 degC and SIL-2 valve XV-

Step 2: Fill Decomposer Reactor D-2304 and Cooler E-2307 loop to 55% level (LT-1301) with pre-mixed Phenol/Aceton

Step 3: Establish tempered cooling water flow through E-2307 via TV-1302 and warm circulation loop to 75.0 degC.

Step 4: Open Sulfuric Acid Catalyst Valve XV-1305 to establish 180 ppmw H2SO4 baseline in D-2304 BEFORE introduci

Step 5: Slowly ramp Concentrated CHP feed (Stream S229 from P-2301A) in 10% increments while monitoring D-2304 ex

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Step 1: Ramp down Concentrated CHP feed (Stream S229) from V-2301 / P-2301A to zero while maintaining D-2304 circ

Step 2: Close Sulfuric Acid Catalyst Valve XV-1305 and flush feed nozzles with pure Cumene.

Step 3: Cool D-2304 loop to < 45 degC via E-2307 before stopping Circulation Pump P-2304A.

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When D-2304 temperature reaches 92.0 degC (TAH-1301):

- Board Operator Action: Immediately verify TV-1302 is 100% open and reduce P-2301A CHP feed rate by 50%.

When D-2304 temperature reaches 115.0 degC (TXSHH-1301 2oo3 voting) or Pressure reaches 5.5 kg/cm2g (PAHH-1301):

- Automatic SIL-2 Interlock I-2304 Action:

1. Closes Sulfuric Acid Valve XV-1305 (< 2 sec).
2. Stops CHP Feed Pump P-2301A/B.
3. Opens Emergency Cumene Quench Valve XV-1309 (Stream S387), flooding D-2304 with cold Cumene to drop temperature.
4. Keeps Circulation Pump P-2304A running and E-2307 Cooling Valve TV-1302 100% open to maximize heat removal.

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- Notice to Operations & Process Safety Engineers:

- * Authoritative Process Data Sheet DS-D2304 specifies D-2304 Internal Design Pressure = 11.0 kg/cm2g (matching
- * P&ID Drawing PID-23-0013 title block carries a conflicting annotation of 12.2 kg/cm2g.
- * Operating limit governance strictly enforces the lower 11.0 kg/cm2g MAWP rating per Corporate Standard STD-EN